F1 Steering Wheel Interface: User Guide

The F1 Steering Wheel Interface is designed for high-speed racing conditions where drivers must process information instantly while staying fully focused on the track. The system combines real-time telemetry, adaptive race data, and minimal cognitive load to help answer three key questions at all times: “Am I fast? Am I safe? What should I do next?”

The interface is divided into several functional zones. At the center of the wheel is the main display area, which contains the most critical driving information. The speed indicator is positioned in the exact center using large white typography on a dark background for maximum readability. The gear indicator is displayed as a large single character above or beside the speed, making it instantly recognizable even in peripheral vision.

At the top edge of the wheel is the RPM and shift light strip. The LEDs transition from green to yellow and then red as engine RPM increases. Green indicates optimal performance, yellow warns the driver that the limit is approaching, and red signals the need to shift immediately. These lights are designed to be understood without direct focus.

The left side of the interface focuses on vehicle performance. Tire condition is shown using four simplified tire indicators with a color-based system: green for optimal, yellow for overheating or wear, and red for critical condition. Fuel and battery systems are displayed using vertical segmented bars instead of raw numbers, allowing faster interpretation during racing.

The right side of the interface focuses on race context and timing. The lap delta display shows whether the driver is gaining or losing time, using green for faster laps and red for slower performance. This section also contains race flags and alerts. Yellow represents caution conditions, blue indicates a faster car approaching, red signals critical issues, and white is used for race control information.

The display adapts depending on the race situation. In qualifying mode, the interface prioritizes lap delta, sector timing, and tire temperature. During race mode, it focuses on fuel management, tire wear, race gaps, and strategy. Under safety car conditions, the display highlights restart timing and tire preparation. Transitions between modes are smooth and consistent to avoid distracting the driver.

A minimal mode is activated during high-speed corners or moments of intense focus. In this mode, only the most essential information remains visible, including speed, gear, RPM lights, and critical alerts. Secondary telemetry is temporarily hidden to reduce visual overload.

The alert system follows a layered structure. Minor issues are shown through subtle color changes, while more serious warnings use blinking icons and stronger visual emphasis. Critical failures trigger dominant red alerts that immediately demand the driver’s attention. To prevent overload, only one major critical alert is displayed at a time.

The interface uses a universal color system throughout the design. Red always indicates critical danger, yellow represents warnings, green signals optimal performance, and blue is reserved for system information. Consistent color usage helps the driver react instantly without needing to read text.

The physical controls are designed for blind interaction. Rotary dials are placed symmetrically and control systems such as brake balance, differential settings, and energy deployment. The dials use distinct textures so they can be identified through touch while wearing racing gloves.

Push buttons are grouped by function. Communication controls include radio activation and pit confirmation, while performance controls include DRS activation, overtake mode, and energy boost systems. Buttons are spaced carefully to prevent accidental presses during aggressive driving.

Every interaction provides immediate feedback through animations, color changes, or illuminated controls. This reassures the driver that the command has been successfully executed without requiring additional attention.

The steering wheel also integrates with the project's AR ecosystem. Pit crew members using AR glasses can visualize live telemetry, race strategy overlays, and driver perspective streams, while smartwatch integration provides synchronized race alerts and team notifications in real time.

Overall, the steering wheel interface is designed to maximize readability, minimize cognitive load, and support split-second decision-making under extreme racing conditions.